LODE.ai / Atlas.ai

PROTOTYPE 0.1 — SCOPE & OPERATING PARAMETERS

Release candidate: v0.10.2-beta **Status:** Controlled prototype test candidate

Intended audience: invited civil/structural engineering testers and academic reviewers

IMPORTANT LIMITATION
This is an academic research prototype. It is not a certified design package, a replacement for a structural engineer, or an executable implementation of the complete BS 8110 standard. It is not approved for construction, tender or statutory submission.

1. Purpose

LODE.ai Prototype 0.1 explores a multimodal assistant for reinforced-concrete engineering workflows. Atlas.ai combines a hosted language/vision model, drawing-text extraction, retrieval from supplied BS 8110 research datasets, selected preliminary calculation modules and an engineer-review workflow.

The prototype is intended to gather structured feedback on usability, retrieval relevance, conversational assistance, drawing-observation quality and selected calculation workflows. It is not a certified design package or an autonomous engineering authority.

2. Intended operating domain

- **Primary domain:** preliminary reinforced-concrete building design assistance.
- **Reference basis:** supplied BS 8110-related Q&A and beam-design-rule JSON datasets.
- **AI provider:** Groq-hosted model endpoint, configured through server-side `GROQ_API_KEY` and `LODE_MODEL` settings.
- **Drawing text extraction:** Tesseract OCR, where the Tesseract executable is installed.
- **Application:** Python / Streamlit browser interface.
- **Project state:** active application session; no persistent multi-user project database.

The configured model identifier is an environment setting, not a guarantee of continued provider availability. Confirm model access before each controlled test session.

3. Functional capabilities included

3.1 Atlas.ai Engineering Adviser

- Accepts natural-language structural engineering and RC design questions.
- Uses recent conversation history within the active session and user-supplied project notes.
- Uses drawing observations as project context only after an engineer marks them verified.
- Retrieves potentially relevant records from bundled knowledge datasets and can expose record references.
- Can explain general engineering concepts and preliminary methods, subject to model and source limitations.

3.2 Drawing Studio

- Accepts PDF and common raster drawing formats supported by the interface.
- Extracts embedded PDF text where available and can run Tesseract OCR on a selected page/image.
- Can send the selected image and OCR text to the configured vision model.
- Presents proposed observations and approximate regions for review; proposals are not automatically verified.

3.3 Engineer Verification

Reviewers can edit, confirm, reject or request clarification on extracted observations. Only entries marked *Verified by engineer* are eligible for Atlas project context. The application does not authenticate the reviewer or independently certify the correctness of a confirmation.

3.4 Analysis and preliminary design modules

- Simply supported beam statics for full-span uniformly distributed load and downward point loads.
- Limited preliminary rectangular RC beam design for the implemented simply supported / UDL case.
- Limited preliminary one-way slab design using a one-metre design strip and simply supported assumptions.
- Limited short tied rectangular column calculation for concentric axial compression, subject to user confirmation of assumptions.
- Preliminary beam reinforcement bar selection/schedule from a supplied required steel area.
- Selected arithmetic checks and equilibrium residual checks.

A passed arithmetic check is not a code-compliance check. Read each module's stated assumptions and limitations before interpreting results.

4. Required inputs and user responsibilities

- Correct project description, design basis, units and applicable standard/edition.
- Geometry, spans, support conditions, load cases and combinations.
- Concrete/reinforcement properties, cover, member dimensions and all module-specific inputs.
- Drawing scale, page selection, image legibility and interpretation of extracted labels/dimensions.
- Confirmation that a selected module's structural assumptions apply to the real structure.

If key inputs are missing, ambiguous or contradictory, Atlas should request clarification or state the limitation rather than silently invent values.

5. Knowledge and citation parameters

- Retrieval searches bundled JSON datasets; it does not automatically query or validate the complete published BS 8110 text.
- Retrieved entries are research-dataset records. Their presence does not prove wording, values, clause reference, amendment status or applicability.
- Verify every clause, factor, limit, equation and design requirement against an authorised copy of the complete applicable standard and amendments.
- If no relevant record is retrieved, Atlas should say so and must not fabricate a clause reference.
- General engineering explanations must be distinguished from statements attributed to retrieved dataset records.

6. Drawing interpretation parameters

- OCR is text recognition, not structural interpretation.
- Vision output is a model proposal, not ground truth.
- Model-reported certainty is qualitative and uncalibrated; it is not a probability of correctness.
- Bounding boxes are approximate proposals, not measurement-grade geometry.
- Dimensions, reinforcement marks, member labels, grids and element classifications require comparison with the original drawing and engineer confirmation.
- No automatic verified building geometry, connected analytical model, finite-element model or load-path model is produced.

7. Explicit exclusions

- Complete/certified BS 8110 compliance checking.

- Full building/global frame or finite-element analysis, load-path analysis, or automatic structural modelling from drawings.
- Full continuous, doubly reinforced or flanged beam design; comprehensive torsion, shear, deflection and crack-width checks.
- Full two-way slab, punching shear, foundation, retaining-wall, slender-column or biaxial-column design.
- Construction-ready detailing, anchorage/lap/curtailment validation, certified bar bending schedules or production DXF drawings.
- Steel design, BEME/BOQ automation, fine-tuning/LoRA, user accounts, role-based access, persistent cloud database or multi-user synchronisation.
- Signed engineering approval, professional certification or construction suitability.

8. Data handling and deployment precautions

- Text and selected drawing images are sent to the configured Groq-hosted inference endpoint when an AI action is used.
- Do not submit confidential, personal, security-sensitive or client-restricted drawings without owner authorisation and review of provider terms and organisational policy.
- Keep the API key in local environment configuration or deployment-platform secrets. Never commit it to source control or expose it in browser code.
- A shared deployment key can consume the owner's quota and incur costs. Broad public access requires authentication, rate/usage controls, abuse prevention, monitoring and an appropriate privacy notice.
- Session state is not a durable record system. Export important test records and store them securely.

9. Conditions before inviting external testers

- Install dependencies in a clean supported Python environment and run the automated test suite.
- Run Streamlit locally and smoke-test every page and principal workflow.
- Verify Groq credentials and configured model using a live connection test.
- Verify Tesseract installation and test OCR on representative drawings.
- Confirm that source code, tracebacks, secrets and implementation payloads are not exposed in ordinary workflows.
- Use non-confidential sample drawings and independently checked calculation examples.
- Explain the limitations to every tester; do not treat outputs as approved design.

A successful software test or live model response does not demonstrate engineering accuracy. Drawing and calculation accuracy require benchmark sets with ground truth and independent engineering review.

10. Suggested tester evaluation areas

Area	What to observe
Usability	Navigation, labels, feedback, error recovery and task completion time
Conversation	Context retention, follow-up understanding, missing-input questions and clarity
Knowledge retrieval	Relevance, exact source traceability, false matches and unsupported questions
Drawing OCR	Exact text accuracy, dimension transcription errors and missed annotations
Vision interpretation	Element-label accuracy, classification errors, uncertainty and box overlap against labelled references
Calculation modules	Input validation, reproducibility and agreement with independently checked examples
Reliability	Crashes, timeouts, provider failures, session loss and diagnostic usefulness

11. Release statement

LODE.ai Prototype 0.1 is suitable only for controlled evaluation within the capabilities and limitations stated in this document. It is not approved for construction, tender, statutory submission or independent professional design. Every engineering input, assumption, retrieved provision, calculation and drawing interpretation must be independently checked by a qualified structural engineer against the complete applicable standards and project requirements.